

↓ 3 R as Philosophical Pillars: Rationalism, Operational Realism, Relationalism ↓			
Method: ↓(Epistemic Hygiene+Relational Origin+Ontological Minimalism+Math Transparency)↓			
Standard Primitives:		$\underbrace{\text{background manifold} + \text{metric}}_{\text{spacetime structure (not derived)}} + \underbrace{\text{G, c, M, fields}}_{\text{energy dynamics (postulated)}}$	
↓ (Ontological reduction by Relational Origin and Minimalism Principles no background allowed) ↓			
One Primitive:		$WILL \equiv SPACE-TIME-ENERGY \implies \boxed{\text{SPACETIME} \equiv \text{ENERGY}}$	
↓ All physically meaningful quantities are relational features of WILL ↓			
Topology of WILL: Relational Closure + Causal Continuity + Isotropy Theorems			
↓ Geometry of WILL: S^1 (1-DOF kinematic Push) and S^2 (2-DOF potential Pull) ↓			
Relational Conservation Theorem:		$\underbrace{\text{Amplitude}^2}_{\text{External Transformation}} + \underbrace{\text{Phase}^2}_{\text{Internal Change}} = 1$	
$\underbrace{1/\beta_Y = 1 + z_\beta}_{\text{transverse Doppler shift}}$	$\underbrace{\beta^2 + \beta_Y^2 = 1}_{\text{Kinematic Push } S^1}$	Relational Projections	$\underbrace{\kappa^2 + \kappa_X^2 = 1}_{\text{Potential Pull } S^2}$ $\underbrace{1/\kappa_X = 1 + z_\kappa}_{\text{gravitational redshift}}$
↓ (by DOF-Indifference Lemma: equal quadratic weight to each independent DOF) ↓			
Closure Theorem:		$\underbrace{1 \text{ PULL}^2}_{\text{Amplitude on } S^2} = \underbrace{2 \text{ PUSH}^2}_{\text{Amplitude on } S^1} \implies \boxed{\kappa^2 = 2\beta^2}$	
↓ (by Relational Conservation Theorem derive metric intervals as scaffolding) ↓			
Minkowski:	$\underbrace{\beta^2 + \beta_Y^2 = 1}_{\text{irreducible primitive}}$	$+ \underbrace{x, y, z, t...}_{\text{coordinate inflation}}$	$= \underbrace{c^2 d\tau^2 = c^2 dt^2 - dx^2 - dy^2 - dz^2}_{\text{mathematical scaffolding}}$
Schwarzschild:	$\underbrace{\kappa^2 + \kappa_X^2 = 1}_{\text{irreducible primitive}}$	$+ \underbrace{x, y, z, t...}_{\text{coordinate inflation}}$	$= \underbrace{c^2 d\tau^2 = c^2(1 - (R_s/r)) dt^2}_{\text{mathematical scaffolding}}$
↓ (by Relational Closure and Causal Continuity Theorems: any change must be balanced) ↓			
Energy Symmetry Law:		$E = E_0 \kappa_X / \beta_Y \longrightarrow \Delta E_{A \rightarrow B} + \Delta E_{B \rightarrow A} = 0$	
$\Delta E_{A \rightarrow B} = E_0(\kappa_{X,B} / \beta_{Y,B} - \kappa_{X,A} / \beta_{Y,A}); \Delta E_{B \rightarrow A} = E_0(\kappa_{X,A} / \beta_{Y,A} - \kappa_{X,B} / \beta_{Y,B})$ ↓			
Speed of light: $\beta \rightarrow 1 \rightarrow \beta_Y \rightarrow 0 \rightarrow E = E_0 \cdot \kappa_X / 0 \rightarrow \infty \rightarrow$ requires ∞ energy			
Event Horizon: $\kappa \rightarrow 1 \rightarrow \kappa_X \rightarrow 0 \rightarrow E = E_0 \cdot 0 / \beta_Y \rightarrow 0 \rightarrow$ requires 0 energy			
by Unified Relational Scaling Lemma: Equivalence Principle		$m_g \equiv m_i \equiv m = E_0 / c^2$	
↓ (Lagrangian and Hamiltonian derived as collapsed single-point approximations) ↓			
$\underbrace{\Delta E_{A \rightarrow B} + \Delta E_{B \rightarrow A} = 0}_{\text{exact symmetry law}}$		$\underbrace{\frac{1}{2}(\kappa_A^2 - \kappa_B^2) + \frac{1}{2}(\beta_B^2 - \beta_A^2)}_{\text{first order approximation}} \approx \underbrace{T \pm V}_{\text{ontological collapse}}$	
↓ Relational Orbital Mechanics		R.O.M.: a closed algebraic system of equations ↓	
↓ Schwarzschild Metric $\tau_o^2 = \kappa_{X_o}^2 - (\kappa_{X_o}^2 \beta_R)^2 \kappa_{X_o}^{-2} - (\kappa_{X_o} \beta_T)^2$ as Scaffolding ↓			
$\underbrace{\tau_o^2 = \kappa_{X_o}^2 \beta_{Y_o}^2}_{\text{irreducible primitive}}$	$+ \underbrace{x, y, z, t...}_{\text{coordinate inflation}}$	$= \underbrace{\left(\frac{d\tau}{dt}\right)^2 = (1 - R_s/r) - \frac{(dr/cdt)^2}{(1 - R_s/r)} - \left(\frac{rd\phi}{cdt}\right)^2}_{\text{mathematical scaffolding}}$	
↓(by Factorization Theorem: every quantity X splits into unitless core $\hat{X}$ and scale label S_X)↓			
Unitless Core:		$X = \underbrace{\hat{X}(\beta, e)}_{\text{unitless core}} \cdot \underbrace{S_X(R_s, c, G)}_{\text{units-scale}}$ G and M are unit-conversion factors.	
↓ (by Inverse Square, Closure and Causal Continuity Theorems) ↓			
No Singularities: $\beta_{\max}^2 = 1 \rightarrow \kappa_{\max}^2 = 2 \rightarrow r_{\min} = R_s / \kappa_{\max}^2 \rightarrow r_{\min} = R_s / 2$			
The point $r = 0$ is absent from the admissible relational range			
↓ (derive Relational Field Equation:) ↓			
$\boxed{\kappa^2 = R_s / r = \rho_{\text{field}} / \rho_{\max}}$		$\implies \boxed{\text{Spacetime Geometry}(R_s / r)} \equiv \boxed{\text{Energy Density}(\rho / \rho_{\max})}$	
↓ Theoretical Ouroboros ↓			
$\boxed{\text{SPACETIME} \equiv \text{ENERGY}}$		$\Leftrightarrow \boxed{\text{SPACETIME GEOMETRY} \equiv \text{ENERGY DENSITY}}$	